



Guidelines

Hyperkalaemia – Acute management

Reference #: HRM028

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital-Wide	Medical, Nursing, Pharmacy

Introduction

This guideline provides guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the acute management of hyperkalaemia.

This document supports and complies with related legislation, policy and other related documents [here](#).

Definitions

Hyperkalaemia	Serum potassium >5.2 mmol/L
Pseudo-hyperkalaemia	Falsely elevated serum potassium
Thrombocytosis	Elevated platelet count
Leucocytosis	Elevated white blood cell count
Continuous Cardiac Monitoring	Observation tool that allows continuous ECG monitoring at the bedside or via portable monitor/defibrillator. Refer to Cardiac Monitoring Guideline .

1.0 General information

Hyperkalaemia is defined as a serum potassium (K^+) **>5.2 mmol/L**.

Hyperkalaemia is **usually asymptomatic**, but clinical features may include:

- ECG changes or cardiac arrhythmias
- Tingling
- Muscle/limb weakness progressing to flaccid paralysis

Symptomatic hyperkalaemia is a medical emergency that can cause fatal arrhythmias or cardiac arrest

Pseudo-hyperkalaemia is a risk that must be considered and excluded ([see below](#)).

Treatment of hyperkalaemia must be **individualised** – factors to consider include:

- Rate of change (i.e. acute vs chronic)
- Severity
- Presence of ECG changes and/or symptoms
- History of chronic or end-stage kidney disease

2.0 Pseudo-hyperkalaemia

Pseudo-hyperkalaemia describes a **falsely elevated** serum potassium, due to *in vitro* leakage from cells, during or after blood collection.

Common cause include:

- Haemolysis of sample (due to prolonged transit time to the laboratory, poor/cold storage conditions, difficult venepuncture)
- Thrombocytosis
- Extreme leucocytosis

⚠ Consider pseudo-hyperkalaemia in asymptomatic patients with normal renal function and normal ECG

If you suspect pseudo-hyperkalaemia ➡ Repeat serum K⁺

IF URGENT

Confirm K⁺ with a venous/arterial blood gas
(noting that K⁺ gas levels are generally lower than serum K⁺ levels)

Don't delay treatment if symptoms or ECG changes present

3.0 Common causes of hyperkalaemia¹⁻³

Cause	Mechanism	
Kidney failure	Reduced potassium excretion	
Fluid volume depletion	Reduced potassium excretion secondary to reduced distal tubular water and sodium delivery	
Medication-induced[†]	• ACE inhibitors (i.e. -prils)	• NSAIDs (<i>ibuprofen, diclofenac, celecoxib, naproxen, etc.</i>)
	• ARBs (i.e. -sartans)	• Trimethoprim
	• Potassium-sparing diuretics (e.g. <i>spironolactone, eplerenone, amiloride</i>)	• Calcineurin inhibitors (e.g. <i>ciclosporin, tacrolimus</i>)
	• Potassium supplements	• Pentamidine IV
	• Heparin / Enoxaparin	• Suxamethonium
Increased potassium release from cells	• Metabolic acidosis • Insulin deficiency • Tissue/cell damage (e.g. trauma/crush injury, rhabdomyolysis, tumour lysis syndrome) • Significant GI bleeding leading to ongoing potassium absorption • Large volume blood transfusions	
Hypoaldosteronism	• Primary adrenal insufficiency (i.e. Addison's disease) • Hyporeninaemia (e.g. diabetes, interstitial kidney disease)	

[†] Commonly implicated medications; not an exhaustive list.

4.0 ECG and Cardiac monitoring⁴

The **ECG** is the most readily available diagnostic tool to assess cardiac toxicity and risk of arrhythmias in hyperkalaemic patients.

Continuous cardiac monitoring allows for early recognition of life-threatening arrhythmias.

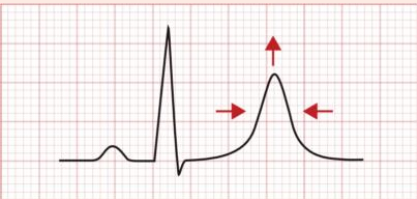
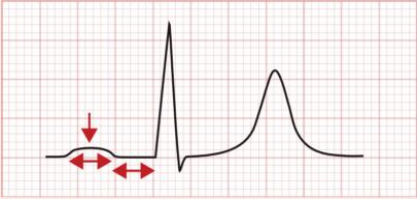
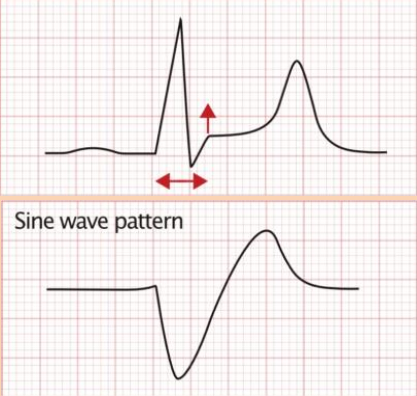
- ⇒ All patients with a serum **K⁺ ≥ 6.0** mmol/L require an **urgent 12-lead ECG**
- ⇒ All patients with a serum **K⁺ ≥ 6.5** or **ECG changes** require **continuous cardiac monitoring**

Hyperkalaemia causes arrhythmias by causing hyperpolarisation of cells, making them less able to depolarise when necessary.

- ECG features are only present in **~66% of patients** with **severe hyperkalaemia** (K⁺ ≥ 6.5 mmol/L).
- Consider the clinical picture alongside ECG – rate of rise and severity of hyperkalaemia, as well as acuity of patient.

Arrhythmias can occur at any time without prior ECG changes

Table 1: Potential ECG changes with increasing severity of hyperkalaemia

Serum Potassium	Potential ECG abnormalities	ECG
5.5 – 6.5	• Peaked T waves	
6.5 – 8.0	• Peaked T waves • Prolonged PR interval • Flat / absent P waves • Widening of QRS complex	
> 8.0	• Absence of T wave • Bradyarrhythmias ○ Sinus bradycardia ○ AV block ○ Slow atrial fibrillation • Sine wave (pre-terminal) • VF, asystole	

5.0 Management of hyperkalaemia

Commence management as per [Appendix 1 \(flowchart\)](#) and [Appendix 2 \(table\)](#).

Emergency treatment for life-threatening hyperkalaemia of any cause, is to:

- Administer IV calcium (immediate cardiac stabilising effect).
- Correct any volume depletion to optimise kidney perfusion.

6.0 Management of hyperkalaemia in Special Patient Populations

6.1 Acute anuric/oliguric kidney failure

- Commence management as per [Appendix 1](#) and [Appendix 2](#).
- Seek urgent Renal advice – may require dialysis.

6.2 End stage kidney failure (Stage 5 CKD)⁴

- Urgent treatment generally not required in mild-to-moderate (5.2 – 6.4 mmol/L) chronic hyperkalaemia – consider:
 - Medication review ([see above](#); nonselective beta blockers such as propranolol or labetalol can cause significant hyperkalaemia in ESRF)
 - Dietary modifications
 - Diuretics (may be less effective in oligo-anuric patients)
 - Potassium exchange resins/polymer
- Seek Renal advice in **severe** (≥ 6.5 mmol/L) and persistent hyperkalaemia – dialysis may need to be expedited.
 - Commence management as per [Appendix 1](#) and [Appendix 2](#).
 - Important to continue close monitoring of serum potassium level even after initial treatment.

6.3 OPH patients

- Commence management as per [Appendix 1](#) and [Appendix 2](#).
- The decision to transfer to a SCGH cardiac monitored bed should be made by the most senior medical staff available.

6.4 Adrenal insufficiency and hypoaldosteronism^{1,5}

- Corticosteroid replacement is the main treatment - insulin should be avoided.

6.5 Acute digoxin toxicity^{6,7}

- Withhold digoxin
- If serum **K⁺ > 6.0 mmol/L** consider giving Digoxin specific antibody fragment (DigiFab[®]) – the Clinical Toxicology service at SCGH can give management advice (contactable 24/7 via the hospital switchboard).
 - NB: DigiFab[®] is not available at OPH.
- Commence hyperkalaemia management as per [Appendix 1](#) and [Appendix 2](#).
 - NB: Calcium was previously avoided when managing hyperkalaemia in patients with digoxin toxicity, but the risk of harm may have been overstated.

6.6 Diabetic Ketoacidosis (DKA)

- Refer to [SCGH Adult DKA Guideline](#)

6.7 Hyperosmolar Hyperglycaemic State (HHS)

- Refer to [SCGH Adult HHS Guideline](#)

Compliance, monitoring and evaluation

- Use of Clinical Incident Management Systems (CIMS) to review trends and investigate incidents as required.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standard/s:

- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Formulary Management Procedures MMP013](#)
- [SCGH Hospital Policy 141 Medicines Management](#)
- NSQHS Standard 1, 4, 5, 6, 8

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline 51: Medication Management](#)

References

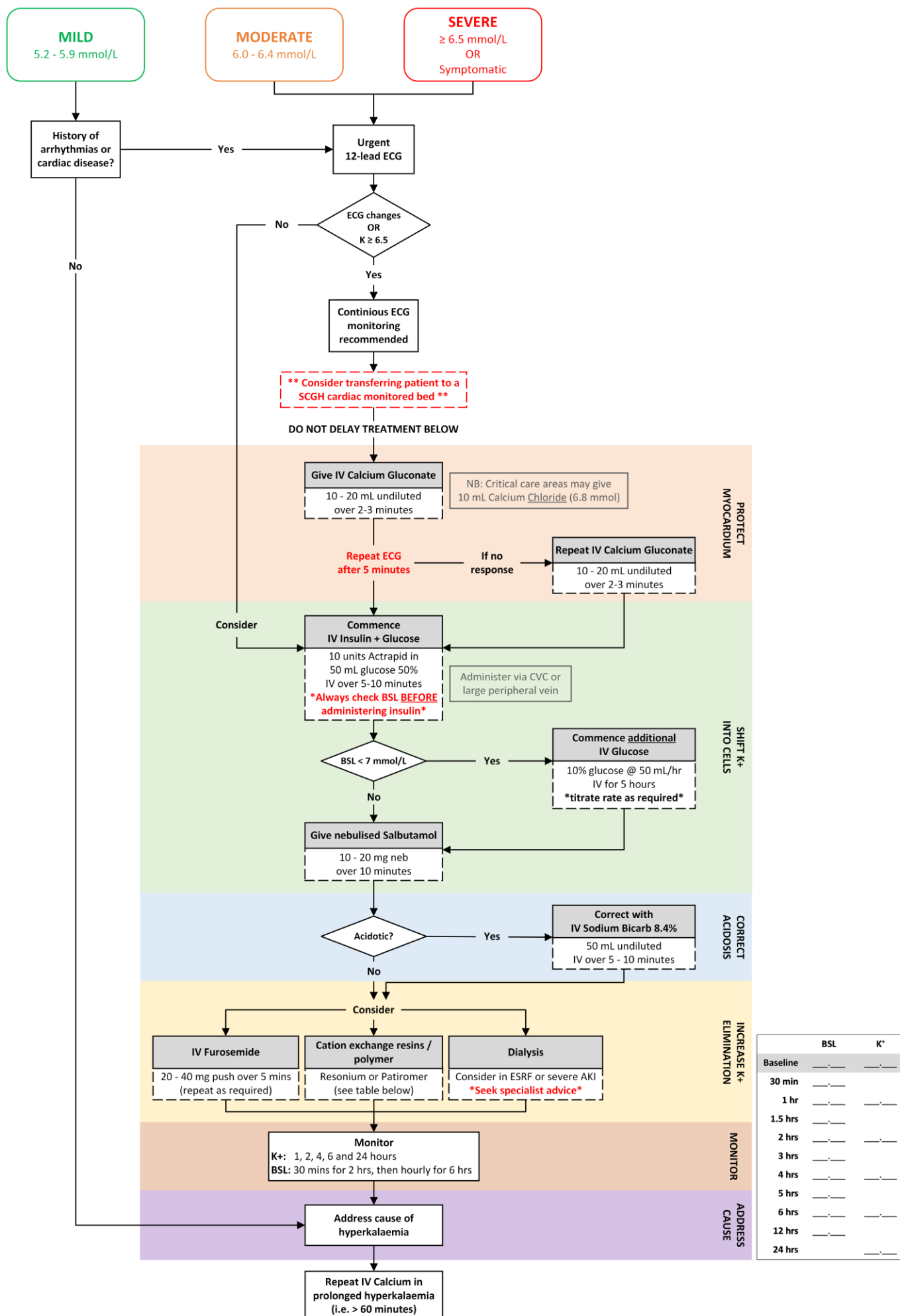
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Authorisation & Version Control

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.



Appendix 2: Medicine administration guide

Treatment	Dose & Administration	Onset / Peak / Duration	Approximate Potassium Reduction	 Precautions	Comments
Protect Myocardium					
IV Calcium Gluconate (preferred) OR IV Calcium Chloride (critical care)	10 to 20 mL (2.2 to 4.4 mmol Ca ²⁺) Inject undiluted into a large vein over 2-3 minutes OR 10 mL (6.8 mmol Ca ²⁺) Inject undiluted into a large vein over 2-3 minutes	Onset: 1-5 minutes Peak: N/A Duration: 30-60 minutes	DOES NOT LOWER POTASSIUM	High risk medication – fatal in overdose or rapid administration • Highly irritant - extravasation can cause tissue necrosis. Monitor injection site closely. • Flush well with sodium chloride 0.9% or glucose 5% to avoid vein irritation.	NB: Calcium Chloride is 3x more potent than Calcium Gluconate • Calcium Gluconate is less irritating to the skin and generally preferred. • Calcium Chloride may be used in critical care areas. • IV calcium is effective within 3 minutes at improving adverse ECG appearances – the dose should be repeated if there is no effect within 5-10 minutes. • Duration of action only 30-60 minutes, so further doses may be needed if hyperkalaemia remains uncontrolled.
Shift K⁺ into Cells					
IV Insulin + Glucose 50%	10 units Actrapid® mixed in 50 mL Glucose 50% Administer via CVC or large peripheral vein over 5 - 10 minutes	Onset: 10-20 minutes Peak: 30-60 minutes Duration: 4-6 hours	0.5 - 1.5 mmol/L	Check BSL before administering insulin • Up to 75% of patients develop hypoglycaemia within 1-hour. • Glucose 50% may cause vascular irritation when given peripherally.	• Risk factors for hypoglycaemia: ◦ Low pre-admission BSL, insulin naïve, female, abnormal renal function, low body weight, adrenal insufficiency. • Glucose requirements vary. Supplemental glucose may be unnecessary if BSLs are elevated. • Alternative administration (in less urgent settings): 10 units Actrapid® in 250 mL Glucose 10% - Give IV over 15 minutes
Salbutamol (nebulised)	10 – 20 mg nebulised over 10 minutes	Onset: 15-30 minutes Peak: 90-120 minutes Duration: 2-6 hours	Variable: ~0.5 - 1 mmol/L (dose-dependent)	• Do NOT use as monotherapy – variable effects and may cause transient increase in serum K⁺ in the first 15 mins. • IHD / Tachyarrhythmias – may worsen.	• Variable effect: ◦ 20-40% of patients will produce a sub-maximal response. ◦ Up to 40% of ESRF patients do not respond to salbutamol. • Combination with insulin-glucose is more effective than either treatment alone. • May cause mild hyperglycaemia (2-3 mmol/L increase) which may partly protect against insulin-induced hypoglycaemia. • Less effective in patients taking non-selective β-blockers.
Correct Acidosis (if present)					
IV Sodium Bicarbonate 8.4%	50 mL undiluted via large vein over 5-10 minutes Repeat in 1-2 hours if required	Variable	Inconsistent	• Highly irritant - extravasation can cause tissue necrosis. Monitor injection site closely.	• Inconsistent effects – do not use as monotherapy. • Do not give via same IV line as calcium. • Be careful in patients with volume overload.

Treatment	Dose & Administration	Onset / Peak / Duration	Approximate Potassium Reduction	 Precautions	Comments
Increase K⁺ elimination					
IV Furosemide	20 – 40 mg push over 5 minutes Repeat according to urine output and blood pressure	Variable	Variable	Do not use if anuric (ineffective) or severe volume depletion.	May be a useful adjunct; do not use alone for treatment of acute hyperkalaemia.
Cation Exchange Resins	Sodium Resonium® (preferred) PO: 15 g TDS-QID PR: 30-50 g daily (as a retention enema)	Onset: >4 hours Peak: N/A Duration: Variable	Inconsistent + Slow onset	CONTRAINDICATED in bowel obstruction or ileus. Oral: the resin might reduce the absorption of other medicines taken by mouth.	- Consider using Calcium Resonium® in patients with hyponatremia (same dosing). PO: mix with 3–4 mL of water per g of resin (usually 15 g in 50–100 mL). NG: As per PO; mix well and give via enteral feeding tube. PR: mix with 5 mL water or glucose 10% solution per g of resin (usually 30 g in 100–150 mL). - Enema should be retained for at least nine hours if possible. - Do not mix with fruit juices as they contain potassium.
Cation Exchange Polymer	Patiomer (requires IPA for non-CKD 5 patients) PO: 8.4 – 25.2 g daily	Onset: 4-7 hours Peak: N/A Duration: Variable	Lacking strong evidence in the acute setting + Slow onset	Patiomer may reduce GI absorption of some oral drugs given within 3 hours (e.g. mycophenolate mofetil, telmisartan, bisoprolol, carvedilol, nebivolol, levothyroxine, metformin - see product information for full details).	- Slow onset of action, therefore limited effect in rapid control of potassium. PO: The dose should be poured into a glass containing ~40mL of water and then stirred. Another ~40mL of water should be added, and the suspension stirred again thoroughly. The powder will not dissolve and the mixture will look cloudy. Add more water to the mixture as needed for desired consistency.